

A Krawtchouk characterization of PPT fermionic radial channels

Candidate full-solution packet

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Status. Candidate full solution of the PPT-characterization subproblem; likely valid; human review recommended.

1 Source question

Arhancet studies radial quantum channels on even-dimensional fermion algebras and computes their entanglement-assisted classical capacity. Section 8 of the final arXiv version asks for the missing PPT and entanglement-breaking characterizations [1] (PDF page 36):

8 Future directions and open questions

The exact computation of the classical capacity, the quantum capacity and the private capacity of radial multipliers is an interesting open question. A characterization of these quantum channels which are entanglement breaking channels or PPT is missing. Consequently, we intend in a sequel to delve into various properties of these quantum channels and explore variations and generalizations of these multipliers.

Declaration of interest None.

Competing interests The author declares that he has no competing interests.

This packet completely answers the PPT part. It also gives the full partial-transpose spectrum, the exact negativity, and the exact PPT time of the fermionic Ornstein–Uhlenbeck semigroup. Entanglement breaking and the unassisted capacities remain open in general.

2 Definitions and proof intuition

Let $n = 2k$, $N = 2^k$, and let $s_1, \dots, s_n$ be self-adjoint unitaries generating M_N and satisfying $s_i s_j = -s_j s_i$ for $i \neq j$. For $A = \{i_1 < \dots < i_r\}$ put $s_A = s_{i_1} \cdots s_{i_r}$. A radial multiplier has the form

$$R_\phi(s_A) = \phi(|A|)s_A, \quad \phi : \{0, \dots, n\} \longrightarrow \mathbb{C}.$$

We assume that R_ϕ is a quantum channel. The source proves that this is equivalent to $\phi(0) = 1$ and positivity of the associated hypercube density [1, Section 5].

For $0 \leq r, c \leq n$, write

$$K_r(c) = \sum_j (-1)^j \binom{c}{j} \binom{n-c}{r-j}; \quad \sum_{r=0}^n K_r(c) z^r = (1-z)^c (1+z)^{n-c}.$$

These are the binary Krawtchouk polynomials.

Proof intuition. Correcting each Clifford monomial by a simple power of i gives a Hermitian Pauli basis. The normalized Choi state is diagonal in the associated Bell basis. Partial transpose removes one quadratic Clifford sign, while the Bell eigenvalue contributes the same quadratic sign together with the commutation character. On summing over all subsets of a fixed size, that character sum is exactly a Krawtchouk polynomial. Radiality therefore collapses an N^2 -dimensional eigenvalue problem to $n + 1$ scalar inequalities.

3 Complete PPT characterization

Theorem 1 (signed Krawtchouk spectrum). *Let $n = 2k$ and suppose that $R_\phi : M_N \rightarrow M_N$, $N = 2^k$, is a fermionic radial quantum channel. Define*

$$\mu_c = 2^{-n} \sum_{r=0}^n \phi(r) (-1)^{\binom{r}{2} + rc} K_r(c), \quad 0 \leq c \leq n. \quad (1)$$

Then the partial transpose of the normalized Choi state of R_ϕ has eigenvalue μ_c with multiplicity $\binom{n}{c}$. Consequently,

$$R_\phi \text{ is PPT} \iff \mu_c \geq 0 \quad (0 \leq c \leq n).$$

Moreover, its negativity and logarithmic negativity are respectively

$$\mathcal{N}(R_\phi) = \frac{1}{2} \left(\sum_{c=0}^n \binom{n}{c} |\mu_c| - 1 \right), \quad E_{\mathcal{N}}(R_\phi) = \log_2 \left(\sum_{c=0}^n \binom{n}{c} |\mu_c| \right).$$

Proof. For $A \subset [n]$, put $q(A) = \binom{|A|}{2}$ and

$$u_A = i^{q(A)} s_A.$$

The reversal of the $|A|$ Clifford generators shows that u_A is a self-adjoint unitary. The u_A form a Hilbert–Schmidt orthogonal basis of M_N , with $\text{Tr}(u_A u_B) = N \delta_{A,B}$.

Let Θ be the complex-linear $*$ -antiautomorphism that fixes every s_i and reverses products. Every $*$ -antiautomorphism of M_N is a unitary conjugate of ordinary matrix transpose. Thus the Θ -adapted Choi construction below differs from the standard one only by a local unitary and has the same PPT spectrum. Reversing s_A gives

$$\Theta(u_A) = (-1)^{q(A)} u_A. \quad (2)$$

Let P_Θ be the Θ -adapted normalized maximally entangled projector. Its Pauli-basis expansion is

$$P_\Theta = N^{-2} \sum_{A \subset [n]} \Theta(u_A) \otimes u_A.$$

Indeed, if $\Theta = \text{Ad}_W \circ (\cdot)^T$, then $P_\Theta = (W \otimes I) |\Omega\rangle\langle\Omega| (W^* \otimes I)$ for the standard maximally entangled vector Ω . Thus the corresponding Choi state is locally unitarily equivalent to the standard one. Applying Θ on its first leg is exactly the standard partial-transpose test in these adapted coordinates: $\Theta^2 = \text{id}$ implies that $W\bar{W}$ is scalar, and a direct calculation gives $(\Theta \otimes \text{id})((W \otimes I)X(W^* \otimes I)) = X^\Gamma$. The normalized Choi state and this partial transpose are therefore

$$\begin{aligned} \rho_\phi &= N^{-2} \sum_A (-1)^{q(A)} \phi(|A|) u_A \otimes u_A, \\ \rho_\phi^\Gamma &= N^{-2} \sum_A \phi(|A|) u_A \otimes u_A. \end{aligned}$$

The operators $u_A \otimes u_A$ commute. Choose a unit vector Ω_Θ spanning the range of P_Θ and, for $C \subset [n]$, define the adapted Bell vector $\Omega_C = (I \otimes u_C)\Omega_\Theta$. These $N^2 = 2^n$ vectors are orthonormal. The adapted maximally entangled vector satisfies $(x \otimes I)\Omega_\Theta = (I \otimes \Theta(x))\Omega_\Theta$. Using this identity and then commuting u_A through u_C shows that

$$(u_A \otimes u_A)\Omega_C = (-1)^{q(A)+\omega(A,C)}\Omega_C, \quad \omega(A, C) = |A||C| - |A \cap C| \pmod{2}. \quad (3)$$

Therefore the eigenvalue indexed by C is

$$2^{-n} \sum_A \phi(|A|)(-1)^{q(A)+\omega(A,C)}.$$

If $r = |A|$ and $c = |C|$, then

$$\sum_{|A|=r} (-1)^{|A \cap C|} = K_r(c).$$

Together with (3), this is precisely (1); it depends only on c , and there are $\binom{n}{c}$ choices of C . Positivity of ρ_ϕ^Γ is now equivalent to the displayed inequalities. The two negativity formulas follow by summing the absolute eigenvalues. $\square$

For completeness, the source's complete-positivity test also becomes a Krawtchouk list. The hypercube density takes the value

$$h_c = \sum_{r=0}^n \phi(r) K_r(c)$$

at any sign vector with c negative coordinates. Hence a symbol defines a radial quantum channel exactly when $\phi(0) = 1$ and $h_c \geq 0$ for every c ; among those channels, Theorem 1 adds the $n + 1$ inequalities $\mu_c \geq 0$ and gives a complete finite characterization of PPT.

4 Exact Ornstein–Uhlenbeck threshold

Corollary 2. *For $0 \leq t \leq 1$, let T_t be the fermionic Ornstein–Uhlenbeck channel with symbol $\phi(r) = t^r$. Then*

$$\mu_c = 2^{-n}(1+t^2)^{n/2} [\cos((n-2c)\theta) + (-1)^c \sin((n-2c)\theta)], \quad \theta = \arctan t. \quad (4)$$

In particular,

$$T_t \text{ is PPT} \iff t \leq \tan \frac{\pi}{4n}.$$

In semigroup time $t = e^{-s}$, the exact PPT time is $s_{\text{PPT}} = \log \cot(\pi/(4n))$.

Proof. The periodic sign satisfies

$$(-1)^{\binom{r}{2}} = \Re((1-i)i^r).$$

Insert this identity into (1) and use the Krawtchouk generating function at $z = i(-1)^c t$. Writing $1 + it = (1 + t^2)^{1/2} e^{i\theta}$ gives (4).

If $\theta \leq \pi/(4n)$, then $|(n-2c)\theta| \leq \pi/4$ for every c , so the bracket in (4) is nonnegative. Conversely, the $c = n$ eigenvalue is negative immediately after $\theta = \pi/(4n)$. There can be no later re-entry: $T_a T_b = T_{ab}$, and local completely positive postcomposition preserves PPT. Indeed, if some T_t with $t > \tan(\pi/(4n))$ were PPT, choose $t' < t$ just above the displayed threshold. Then $T_{t'} = T_{t'/t} T_t$ would be PPT, contradicting the negative $c = n$ eigenvalue of $T_{t'}$. $\square$

5 A rigorous entanglement-breaking region

The exact entanglement-breaking characterization remains open, but the same expansion gives a useful dimension-explicit interval.

Proposition 3. *The Ornstein–Uhlenbeck channel T_t is entanglement breaking whenever*

$$t \leq 2^{1/n} - 1.$$

Consequently, if t_{EB} denotes the endpoint of its entanglement-breaking interval, then

$$2^{1/n} - 1 \leq t_{\text{EB}} \leq \tan \frac{\pi}{4n}.$$

Proof. For every traceless Hermitian unitary u , the normalized states proportional to $I \otimes I \pm u \otimes u$ are separable: expand them using the two spectral projections of u . Hence a Clifford-diagonal Choi state is separable whenever the sum of the absolute values of its nonidentity correlation coefficients is at most one. For T_t this sum is

$$\sum_{r=1}^n \binom{n}{r} t^r = (1+t)^n - 1.$$

This proves the lower bound. Every entanglement-breaking channel is PPT, so Corollary 2 gives the upper bound. $\square$

6 Verification, novelty, and scope

An independent matrix script constructs the Jordan–Wigner representation for $n = 2, 4, 6$ and matches the entire partial-transpose spectrum against (1) to floating-point precision. It separately checks (4). These computations audit signs and normalization but are not part of the proof.

A bounded novelty search on 11 August 2026 used the exact title, the source’s PPT sentence, and combinations of “fermionic radial multiplier”, “PPT”, “Krawtchouk”, “Clifford diagonal channel”, and “fermionic Ornstein–Uhlenbeck entanglement breaking time”. It found the source and general Bell-diagonal literature, but no later solution of this exact characterization or the signed transform above. Novelty confidence is moderate.

The theorem resolves only the PPT part of the final source question. It does not characterize all entanglement-breaking radial channels and does not compute their unassisted classical, quantum, or private capacities. Numerical evidence for a sharper Ornstein–Uhlenbeck entanglement-breaking threshold is kept outside the theorem.

Human-review recommendation. Audit the use of the reversal antiautomorphism as transpose, the sign in (3), and the $N^{-2} = 2^{-n}$ Choi normalization. Then verify the semigroup no-re-entry step in Corollary 2. These are the only places where convention changes can alter the formulas.

References

- [1] C. Arhancet, *Entanglement-assisted classical capacities of some channels acting as radial multipliers on fermion algebras*, arXiv:2402.15440v6 (2024).